

Neural Similarity Search for Drug Target Prediction

Nicolas Madella-Mella - k12021216

Department of Machine Learning

Johannes Kepler University Linz

Austria, Linz 4040

Github-Projekt

`nicolas.madellamella@gmail.com`

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Abstract

Building models for molecular properties and activities with minimal measurements is a crucial challenge in drug discovery. Typically drug discovery projects begin with one or a few known active molecules and then face the task of identifying promising candidates for further screening. Few-shot learning methods have been introduced to computer-aided drug design to enhance this critical phase of the process. It has been shown that effective few-shot learning methods and architectures can significantly reduce time and costs associated with in-vitro testing, thereby accelerating the drug development process. In this work, the Neural Similarity Search was reimplemented, showing that a fully-connected Deep Neural Network using scaled exponential units in a Siamese fashion and paired with Similarity Search is suitable to tackle the problem of low measurements. For this, the SIDER dataset was used to evaluate the model’s performance. The Neural Similarity Search not only significantly outperformed a Random Forest baseline, but also achieved superior results compared to state-of-the-art benchmarks on the SIDER dataset.

1 The story summary

The focus of this report is on utilizing few-shot learning methods to address the challenge of building predictive models with limited data in drug discovery. Initially, a descriptor-based, fully-connected Deep Neural Network (DNN) with scaled exponential units (Klambauer et al., 2017), configured in a Siamese Network architecture (Koch et al., 2015) was employed. This approach was chosen based on the framework suggested by Schimunek et al. (2021). The performance of this Neural Network model was assessed against a Random Forest baseline, using metrics such as the Area Under the Curve (AUC) and the normalized Area Under the Precision-Recall Curve (Δ AUC-PR), as suggested

in Stanley et al. (2021). The evaluation results demonstrated that the neural variant significantly outperformed the Random Forest baseline, indicating the effectiveness of embedding-based few-shot learning methods paired with this type of model architecture and Similarity Search. The superiority is evident even with a minimal number of measurements, as exemplified by the SIDER dataset (Kuhn et al., 2016). Additionally, the Neural Similarity Search model not only exceeded the Random Forest baseline but also achieved superior results compared to state-of-the-art benchmarks.

2 Introduction

Enhancing human health, fighting diseases, and addressing future pandemics, as demonstrated by recent experiences with the COVID pandemic, have shown that discovering new drugs quickly and efficiently is essential. But the drug discovery process is time-consuming and cost-intensive (Arrowsmith, 2011). In order to reduce costly and time-consuming in vitro experiments and accelerate drug development, machine learning algorithms are used to predict the binding affinity of molecules towards target proteins, which allows to narrow down the number of suitable candidates.

Specifically Deep Learning models were used to reach those goals, with the disadvantage of them being very data hungry (Marcus, 2018) and therefore need very large amounts of measurements for accurate activity prediction. So to reach high predictive performance for Deep Learning-based binding activity prediction, hundreds or thousands of measurements per task are needed. Well-performing models for activity prediction tasks of ChEMBL have been trained with an average of 3,621 activity points per task (Mayr et al., 2018). The ExCAPE-DB dataset offers an average of 42,501 measurements per task (Sun et al., 2017). Wu et al. (2018) provided a large-scale benchmark for molecular machine learning, including prediction models for the SIDER dataset (Kuhn et al., 2016) with an average of 5,187 data points, which is also the dataset being focused on in this work, Tox21 (Huang et al., 2016; Mayr et al., 2016) with an average of 9,031 data points, and ClinTox (Wu et al., 2018) with 1,491 measurements per task. However, for many drug targets, the available measurements are very limited (Waring et al., 2015; Altae-Tran et al., 2017; Stanley et al., 2021) due to the resources required for in vitro experiments. Therefore, methods that can utilize few measurements to build valuable property and activity prediction models are necessary.

The challenge of training models with limited data is addressed by the machine learning fields of meta-learning and few-shot learning (Schmidhuber, 1987; Bengio et al., 1990; Miller et al., 2000; Hochreiter et al., 2001; Bendre et al., 2020; Y. Wang et al., 2020). Few-shot learning, initially developed for image datasets (Bendre et al., 2020; Y. Wang et al., 2020), is now being applied to drug discovery (Altae-Tran et al., 2017; Adler et al., 2020; Stanley et al., 2021).

These methods are categorized into four main approaches (Bendre et al., 2020; Y. Wang et al., 2020):

- a Data-augmentation-based approaches generate diverse data points from the limited available samples (Antoniou & Storkey, 2019; Zhao et al., 2019; T. Chen et al., 2020).
- b Embedding-based and nearest neighbor approaches create meaningful representations for low-shot predictions, such as Matching Networks using attention mechanisms (Vinyals et al., 2016) and Prototypical Networks creating class prototypes (Snell et al., 2017).
- c Optimization-based or fine-tuning methods, like MAML, adapt initial weights to new tasks with few optimization steps (Finn et al., 2017).
- d Semantic-based approaches use additional semantic information for new task learning (Li et al., 2019; Schwartz et al., 2022).

In drug discovery, Adler et al. (2020) proposed a cross-domain few-shot learning method using representation fusion. Nguyen et al. (2020) and Guo et al. (2021) explored GNNs combined with meta-learning. Altae-Tran et al. (2017) developed Iterative Refinement Long Short-Term Memory for molecule embeddings. Stanley et al. (2021) created a benchmark dataset for few-shot learning in drug discovery, but connections to classic cheminformatics techniques like Similarity Search (Cereto-Massagué et al., 2015) remain unclear.

Schimunek et al. (2021) therefore presented a framework for embedding-based few-shot learning methods integrating Similarity Search, as well as Deep Learning techniques. This work aims at reimplementing this approach, train it in a few shot manner using the SIDER dataset (Kuhn et al., 2016), which contains a low amount of measurements and compared it to a Random Forest Baseline.

3 Methods

3.1 Description of datasets used:

For this work’s experiments the SIDER dataset was used (Kuhn et al., 2016). It contains information on marketed medicines and their recorded adverse drug reactions. The information is extracted from public documents and package inserts. The available information include side effect frequency, drug and side effect classifications as well as links to further information, for example drug–target relations. SIDER consists of activity measurements from 1427 molecules on 27 tasks, either being inactive or active. As there are no missing values, this leads up to 38529 measurements overall.

3.2 Descriptors and Preprocessing:

In this preprocessing pipeline, molecular data, encoded as SMILES strings, is systematically transformed into a numerical format suitable for multi-task learning. The process begins by converting SMILES strings into RDKit Mol

objects, which accurately represent the molecular structures. Extended Connectivity Fingerprints (ECFPs) (Rogers & Hahn, 2010) are then generated for each molecule, capturing structural features in a high-dimensional vector format. Additionally, RDKit descriptors are computed to quantify various molecular properties. A crucial step involves applying an empirical cumulative distribution function (ECDF), which is first fitted on the molecules of the trainfold and used to normalize the descriptors by transforming raw values into quantiles. This method mitigates the impact of outliers and enhances the consistency of the data across features. The final step involves concatenating the ECFPs and quantile-transformed descriptors, followed by standardization using Scikit-learn’s (Pedregosa et al., 2011) StandardScaler to ensure all features are on a uniform scale. This preprocessing strategy effectively prepares the data for robust and consistent model training, ensuring that molecular features are uniformly processed across all tasks.

3.3 Model architecture:

As introduced by Schimunek et al. (2021), a descriptor-based fully-connected Deep Neural Network with scaled exponential units (Klambauer et al., 2017) as the molecule and memory encoder in a Siamese Network fashion (Koch et al., 2015) was implemented. In the first step, the representations for the query molecule and support set molecules are encoded by this fully-connected network and transformed by the Layer Normalization operation (J. L. Ba et al., 2016). This can be seen as a function h_ω with learnable parameters ω to compute the feature representations for molecules

$$h_{mol}(x) = h_\omega(x)$$

where h_{mol} represents the embedding of molecule x .

Applying this function to the query molecule, as well as to the positive and negative support set molecules respectively, gives the representations q , p_i , n_j , for $i \in \{1, \dots, P\}$ and $j \in \{1, \dots, N\}$ where $N = P = 8$, being the number of support set molecules used in this work’s experiments.

The resulting embeddings are then compared using the dot product to measure similarity. The dot product similarities between the query molecule and each of its corresponding support set molecules are averaged separately for the positive and negative support sets.

$$q_p = \frac{1}{|P|} \sum_{i=1}^{|P|} (q \cdot p_i)$$

$$q_n = \frac{1}{|N|} \sum_{j=1}^{|N|} (q \cdot n_j)$$

Finally, the raw prediction values are multiplied by a constant c , being $1/\sqrt{D}$, where D is the embedding dimension and scaled to a range between zero and one using the sigmoid function.

$$\hat{y} = \sigma(c(q_n - q_p))$$

This process is also illustrated in Figure 1. During the experiments, the cosine similarity was also tested, but no benefits were gained from it, therefore the dot product was being kept as the similarity comparison in the association function.

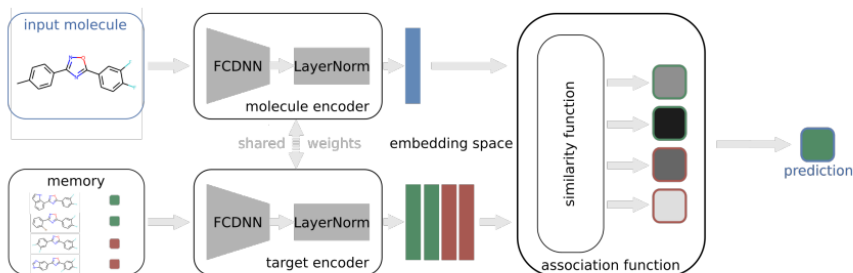


Figure 1: Overview neural search variant

During first attempts at implementing the method, the prototypical network (Snell et al., 2017) was unknowingly mimicked. The key idea of this network is that each class can be represented by a single prototype, which is the mean of the class’s support set in an embedding space. A neural network is used in order to learn a non-linear mapping from input data to this space. Classification of a query point is then done by finding the nearest class prototype.

3.4 Evaluation

The performance of the Neural Similarity Search variant (Schimunek et al., 2021) and the Random Forest baseline are assessed using the Area Under the Curve (AUC), and, as proposed by Stanley et al. (2021), the normalized Area Under the Precision-Recall Curve ($\Delta\text{AUC-PR}$). Each task of the SIDER dataset (Kuhn et al., 2016) is evaluated individually, with the results averaged to provide a comprehensive measure across all tasks. The $\Delta\text{AUC-PR}$ metric quantifies the model’s improvement over random guessing by subtracting the AUC-PR of a random classifier, based on the proportion of positive samples. A positive $\Delta\text{AUC-PR}$ indicates superior performance, whereas a near-zero or negative value suggests performance comparable to or worse than random guessing.

The comparison of the Neural Similarity Search with the two benchmark methods from W. Chen et al. (2022) is limited to the AUC metric, as $\Delta\text{AUC-PR}$ was not reported for those methods.

4 Experiments

4.1 Methods compared

The primary objective of this work is to compare the performance of the Neural Similarity Search model (Schimunek et al., 2021) against a Random Forest baseline. Additionally, a secondary aim was to surpass benchmark results reported in W. Chen et al. (2022), specifically the performance of the Siamese network (Koch et al., 2015), which closely aligns with the architecture used in this study, and the PAR (Property-Aware Relation Network) model (Y. Wang et al., 2021), which previously demonstrated superior results.

The PAR model employs a graph-based molecular encoder to generate initial embeddings, which are subsequently refined into property-aware embeddings via a specialized function that learns an adaptive relation graph, effectively capturing molecular relationships and propagating labels. Initial experiments, conducted using the parameters provided in (Schimunek et al., 2021), yielded promising results. Furthermore, a hyperparameter search script was developed and executed within a defined search space. However, this exploration did not significantly improve upon the evaluations already achieved

4.2 Support sets and Data split

The dataset was divided into training, validation, and test sets, with 60%, 20%, and 20% of the tasks allocated to each split, respectively. During training, for each of the 1,427 molecules, eight active and eight inactive molecules were randomly selected (excluding the original molecule) to form the support set for the training process. In the validation and test sets, a fixed support set of eight active and eight inactive molecules was utilized for each of the remaining 1,411 molecules in the SIDER dataset (Kuhn et al., 2016).

4.3 Hardware Specifications

The experiments were conducted using the following hardware setup:

- **Processor:** AMD Ryzen 5 5600 (3.5 GHz, 6 cores)
- **Memory:** 32 GB DDR4 RAM
- **GPU:** NVIDIA GeForce RTX 4060 TI, 16 GB
- **Operating System:** Microsoft Windows 11 Pro

4.4 Training

The Random Forest baseline consists of 1000 decision trees. Data shuffling was enabled to enhance model robustness against data order effects. The hyperparameters used for the Neural Similarity Search Model (Schimunek et al., 2021) are stated below in Table 1. The model was trained using the Adam optimizer (Kingma & J. Ba, 2014) to minimize binary cross-entropy loss.

To monitor and manage the training statistics of the Neural Similarity Search variant, TensorBoard from the TensorFlow framework (Abadi et al., 2015) was utilized. For both the training and validation sets, the area under the receiver

operating characteristic curve (AUC), as well as the normalized Area Under the Precision-Recall Curve (Δ AUC-PR), was recorded. Additionally, the training and validation loss (binary cross-entropy-loss) were logged and the model with the best performance was consistently saved. Early stopping was used to ensure that the training stops when the model is no longer improving, thus avoiding overfitting and unnecessary computations. The stopping criterion used was Δ AUC-PR, as it provided the best results. To facilitate training across multiple seeds and to average the evaluation results for both the Neural Similarity Search variant and the Random Forest baseline, a separate script was developed and utilized.

Table 1: Hyperparameters used in Training

Hyperparameter	Value
Learning Rate	0.001
Weight Decay	0
Batch Size	32
Input Dimension	2248
Hidden Dimension	1124
Output Dimension	1124
Number of Layers	2
Dropout Probability (p)	0.1

5 Results

In this section, the performance of the trained methods, including the Neural Similarity Search variant (Schimunek et al., 2021) and the Random Forest baseline, are compared. Additionally, benchmark results reported in (W. Chen et al., 2022) are incorporated, specifically those of a Siamese network (Koch et al., 2015), which shares architectural similarities with the Neural Similarity Search variant and the Property-Aware Relation Network (PAR) (Y. Wang et al., 2021), which achieved the best performance on the SIDER dataset (Kuhn et al., 2016) in terms of AUC. A visualization of the training’s impact on the learned embeddings is also presented.

Table 2: Average Performance of Models Across All Seeds (%)

Model	AUC	Δ AUC-PR
Neural Similarity Search	77.97 $\pm$ 2.98	17.79 $\pm$ 3.29
Random Forest Baseline	55.11 $\pm$ 0.966	1.65 $\pm$ 0.585

Table 2 presents the average performance of the Neural Similarity Search and the Random Forest baseline model across all tasks, measured using AUC and Δ AUC-PR. The Neural Similarity Search model significantly outperforms the Random Forest baseline in both metrics.

Table 3: Performance Comparison Across Benchmarks (%)

Model	AUC
Neural Similarity Search	77.97 $\pm$ 2.98
PAR	74.68 $\pm$ 0.31
Siamese Network	71.10 $\pm$ 4.32
Random Forest Baseline	55.11 $\pm$ 0.966

Table 3 compares the AUC performance of the Neural Similarity Search model with several benchmark models, including PAR, Siamese Network and the initial Random Forest baseline. The Neural Similarity Search model demonstrates superior AUC performance compared to all benchmarks.

Table 4: AUC Performance Across Multiple Seeds (%)

Model	Mean $\pm$ Std	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5	Seed 6	Seed 7	Seed 8	Seed 9	Seed 10
Neural Similarity Search	77.97 $\pm$ 2.98	76.15	81.67	76.60	81.06	76.25	83.38	75.30	76.05	75.01	78.20
Random Forest Baseline	55.11 $\pm$ 0.966	53.91	54.81	55.98	55.00	55.21	55.29	56.10	54.23	53.81	56.73

Table 4 details the AUC performance of the Neural Similarity Search and Random Forest models across ten different seeds, representing varied support sets. The Neural Similarity Search consistently outperforms the Random Forest across all seed variations.

Table 5: Δ AUC-PR Performance Across Multiple Seeds (%)

Model	Mean $\pm$ Std	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5	Seed 6	Seed 7	Seed 8	Seed 9	Seed 10
Neural Similarity Search	17.79 $\pm$ 3.29	15.94	16.17	16.03	26.29	15.80	18.81	19.83	16.35	15.76	16.90
Random Forest Baseline	1.65 $\pm$ 0.585	0.89	1.25	2.43	2.00	1.78	1.00	2.54	1.12	1.57	1.96

Table 5 displays the Δ AUC-PR performance of the Neural Similarity Search and Random Forest models across ten seeds, reflecting different support sets. The Neural Similarity Search model consistently achieves higher Δ AUC-PR values compared to the Random Forest.

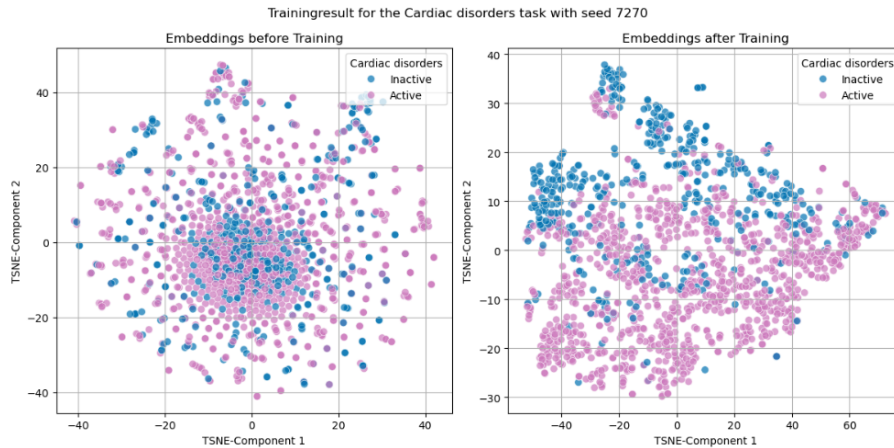


Figure 2: Comparison of Embeddings before and after Training

Figure 2 visualizes the feature embeddings before and after training, illustrating the improvements in clustering and separability as a result of training. The post-training embeddings exhibit clearer groupings, highlighting the impact of the training process on feature representations.

6 Conclusion and Outlook

In this work, the Neural Similarity Search framework (Schimunek et al., 2021) for drug target prediction was reimplemented and was compared against a Random Forest baseline. By applying the model to the SIDER dataset (Kuhn et al., 2016), which contains a relatively small number of samples, it demonstrated the efficacy of few-shot learning methods integrating Similarity Search (Cereto-Massagué et al., 2015) in the drug discovery process. The results show that the Neural Similarity Search model significantly outperformed the Random Forest baseline, achieving higher AUC and Δ AUC-PR scores.

This indicates that embedding-based methods, particularly in a Siamese network configuration (Koch et al., 2015), are more capable of capturing the underlying molecular patterns with limited data compared to traditional machine learning methods. The model’s robust performance across different seeds suggests that it generalizes well, even with variations in the support set, reinforcing the potential of neural architectures for drug discovery when faced with data scarcity.

In comparison to state-of-the-art benchmarks, the Neural Similarity Search model surpassed the performance of both the Siamese Network (Koch et al., 2015) and the PAR model (Y. Wang et al., 2021) reported in W. Chen et al. (2022), highlighting the effectiveness of integrating similarity search mechanisms with deep learning approaches. This success emphasizes the importance of adopting flexible neural architectures that can adapt to the specific demands of drug discovery tasks.

Future work will focus on optimizing the training process with better stopping criteria, as well as incorporating an additional Transformer encoder after the fully-connected network encoder, as proposed by Fifty et al. (2023), building on the existing architecture’s ability to effectively process molecular embeddings. This extended model would allow for deeper contextualization of the relationships between support set molecules and the query molecule.

By adding another Transformer encoder, the aim is to further enhance the model’s capacity to capture complex patterns and dependencies within molecular property prediction tasks. This could enable improved generalization across diverse datasets and boost the model’s performance in few-shot learning scenarios, potentially surpassing current results.

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